

3.1 An Introduction to AS Level Organic Chemistry

Question Paper

Course	CIE A Level Chemistry
Section	3. Organic Chemistry
Topic	3.1 An Introduction to AS Level Organic Chemistry
Difficulty	Medium

Time allowed: 80

Score: /64

Percentage: /100

Question 1a

A chemist is analysing a collection of organic compounds.

Some of these compounds are shown in Table 1.1.

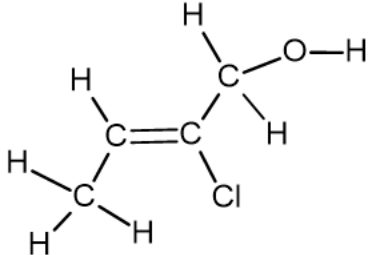
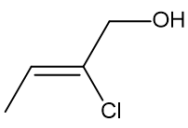
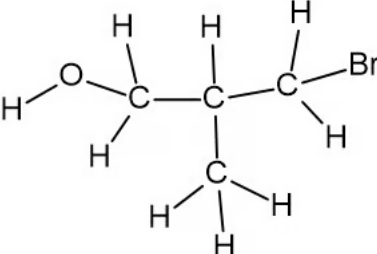
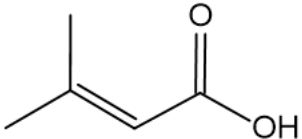
Compound	IUPAC name	Displayed formula	Skeletal formula
1	Z-2-chloro-but-2-en-1-ol		
2	2-chloropropanal		
3			
4			

Table 1.1

Complete Table 1.1.

[3 marks]

Question 1b

State the empirical formula of compound **1**.

[1 mark]

Question 1c

Explain why compound **1** exists as two stereoisomers, but compound **4** does not.

[4 marks]

Question 1d

Compounds **2** and **3** both exhibit the same type of isomerism.

i)

State the type of isomerism shown by compounds **2** and **3**.

[1]

ii)

Draw the **two** isomers of compound **2** to explain how this type of isomerism occurs.

[3]

[4 marks]

Question 2a

Explain how the concept of hybridisation can be used to explain the triple bond present in propyne.

[3 marks]

Question 2b

Consider the molecule shown in Fig. 2.1 which contains both sigma and pi bonds.

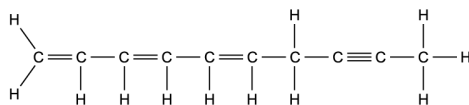


Fig. 2.1

State the number of carbon atoms that exhibit sp² hybridisation in this molecule.

[1 mark]

Question 2c

Complete Table 2.1 to show the hybridisation of the nitrogen atoms in NF_4^+ , N_2H_2 and N_2H_4 .

Table 2.1

	NF_4^+	N_2H_2	N_2H_4
Hybridisation			

[3 marks]

Question 3a

This question is about the structure and reactions of organic compounds.

Fig. 3.1 shows the displayed formula of a type of fuel known as gasoline.

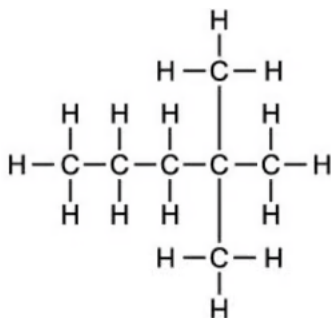


Fig. 3.1

i)

State the IUPAC name of gasoline.

[1]

ii)

Give the general formula for the homologous series to which gasoline belongs.

[1]

[2 marks]

Question 3b

The gasoline shown in Fig. 3.1 can react with chlorine in the presence of UV light to form various chlorinated products.

i)

State the type of organic mechanism used in this reaction.

[1]

ii)

Draw the skeletal formula of the **four** monochlorinated isomers formed when the gasoline in Fig. 3.1 reacts with chlorine.

[4]

iii)

State the type of isomerism shown by the isomers in (ii).

[1]

[6 marks]

Question 3c

There are two possible reactions of 1-bromopropane with sodium hydroxide.

Fig. 3.2 shows the mechanism for the reaction of aqueous sodium hydroxide with 1-bromopropane.

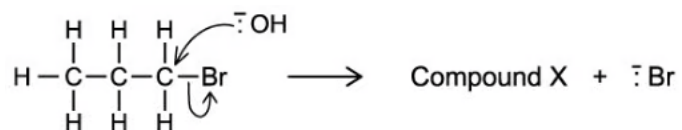


Fig. 3.2

- i)
State the role of the aqueous hydroxide ion in this reaction.

[1]

- ii)
Name compound X.

[1]

- iii)
In different reaction conditions, an alternative reaction occurs between the hydroxide ion and 1-bromopropane to form propene and a small molecule.

State the reaction conditions and name the type of mechanism used in this alternative reaction.

[2]

[4 marks]

Question 3d

Pentanal reacts with hydrogen cyanide ions to form 2-hydroxyhexanenitrile as outlined in Fig. 3.3.

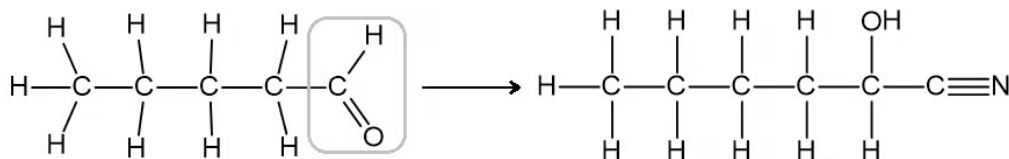


Fig. 3.3

i)

In terms of orbitals and bonding, explain why the shape of the carbonyl carbon highlighted in Fig. 3.3 can be described as trigonal planar.

[3]

ii)

State the shape and bond angle adopted by the carbonyl carbon after the reaction with hydrogen cyanide.

[1]

[4 marks]

Question 4a

There are many different chemicals with the same molecular formulae but different arrangements of atoms in three-dimensional space.

i)

Name the shortest straight-chain alkane to have structural isomers and the isomer.

[2]

ii)

State the type of isomerism shown by the isomers in part (i).

[1]

[3 marks]

Question 4b

C_4H_8 exists as several isomers.

i)

Name the homologous series that C_4H_8 belongs to.

[1]

ii)

Draw the skeletal formulae of **two** cyclic C_4H_8 isomers and **two** C_4H_8 isomers that show cis / trans isomerism.

[4]

iii)

There are **two** unsaturated C_4H_8 isomers that do **not** show cis / trans isomerism. Name these isomers and explain why they do not show cis / trans isomerism.

[2]

[7 marks]

Question 4c

Draw the displayed formulae of the following compounds and, where appropriate, identify any chiral centres.

- 2,4-dimethylhexan-3-ol
- 2,4-dimethyl-3-chloropentane
- 2-aminobutane

[6 marks]

Question 4d

1-bromopropan-1-ol exhibits optical isomerism.

Draw 3D representations of the **two** enantiomers of 1-bromopropan-1-ol.

[2 marks]

Question 5a

Hybridisation of the s and p orbitals can be used to explain the shapes observed in molecules.

Explain why cyclohexane C_6H_{12} has a puckered, non-planar shape by making reference to the C–C–C bond angles and the type of hybridisation of carbon in the molecule.

[4 marks]

Question 5b

Urea, $CO(NH_2)_2$, is present in solution in animal urine.

Describe the hybridisation of C and N in the molecule, giving approximate bond angles.

[4 marks]

Question 5c

Describe the hybridisation of the carbon atom in methane and explain how the concept of hybridisation can be used to explain the shape of the methane molecule.

[4 marks]

Question 5d

A molecule of ethanol is shown in Fig. 5.1.

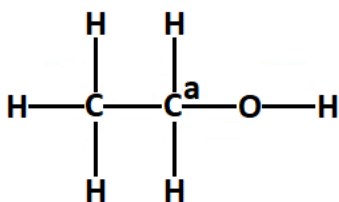


Fig. 5.1

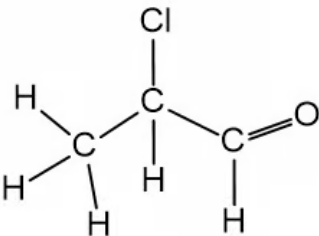
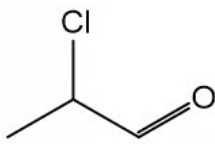
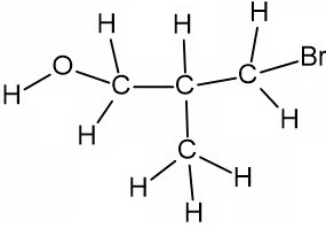
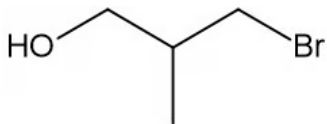
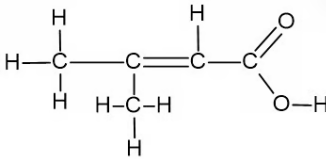
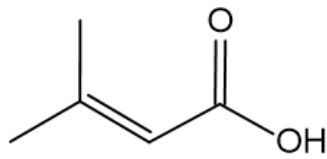
Deduce the hybridisation of the carbon atom marked in the diagram below.

[1 mark]

Model Answers: Medium

1a

a) The IUPAC name, displayed formulae and / or skeletal formulae to complete Table 1.1 are:

Compound	IUPAC name	Displayed formula	Skeletal formula
2	2-chloropropanal		
3	3-bromo-2-methylpropan-1-ol		
4	3-methyl-but-2-enoic acid		

- Compound 2 displayed formula correct
AND
Compound 2 skeletal formula correct; [1 mark]
- Compound 3 IUPAC name correct
AND
Compound 3 skeletal formula correct; [1 mark]
- Compound 4 IUPAC name correct
AND

Compound 4 displayed formula correct; [1 mark]

[Total: 3 marks]

- Compound 2
 - The -prop- portion of the name tells you that the main carbon chain is 3 carbons long
 - The -an- portion of the name tells you that the main carbon chain contains single bonds
 - The -al portion of the name tells you that carbon-1 is a CHO / aldehyde functional group
 - The 2-chloro portion of the name tells you that there is a chlorine attached to carbon-2
 - **Careful:** A common mistake drawing the skeletal formula of any aldehyde is to forget to include the hydrogen atom that is attached to the carbonyl / C=O bond
- Compound 3
 - The main carbon chain is 3 carbons long with single bonds, so the name is propane based
 - There is an alcohol functional group, a methyl functional group and a bromine functional group
 - The alcohol is highest priority, so it is on carbon 1
 - Propan-1-ol
 - The methyl group is the second highest priority
 - 2-methylpropan-1-ol
 - The bromine group is the lowest priority
 - 3-bromo-2-methylpropan-1-ol
 - For the skeletal formula:
 - Re-draw the displayed formula
 - Remove the carbon symbols
 - This will sometimes require you to angle the carbons to ensure that they are clear and obvious
 - Remove all of the hydrogen atoms **except** the hydrogen atom attached to the oxygen
- Compound 4
 - The longest carbon chain is 4 carbons long, so the name is but- based
 - There is a carboxylic acid group, which is automatically carbon-1
 - Butanoic acid

- There is a carbon=carbon double bond on carbon-2
 - But-2-enoic acid
- There is a methyl group on carbon-3
 - 3-methylbut-2-enoic acid
- For the displayed formula:
 - Re-draw the skeletal formula
 - Place carbon symbols on each point / junction in the compound **AND** on the ends
 - Add the appropriate number of bonds to hydrogen atoms to each carbon atom
 - **Careful:** Carbon has a maximum of 4 bonds so take care, particularly around double bonds

1b

b) The empirical formula of compound **1** is:

- C_4H_7ClO ; [1 mark]

[Total: 1 mark]

- The empirical formula of an organic compound gives the simplest possible whole number ratio of the different atoms of each element in a compound
 - The molecular formula of compound **1** can be found by adding up all the different types of atoms of the compound and is C_4H_7ClO
 - Since the molecular formula cannot be further simplified, the empirical formula is in this case the same as the molecular formula

1c

c) Compound **1** exists as two stereoisomers, but compound **4** does not because:

- There is no free rotation about the carbon-carbon double bond / $C=C$; [1 mark]
- Due to the presence of π bonds; [1 mark]
- The (carbon-carbon) double bond / $C=C$ of compound **1** has different functional groups; [1 mark]
- The (carbon-carbon) double bond / $C=C$ of compound **4** has the same functional group / methyl groups on one side / end; [1 mark]

[Total: 4 marks]

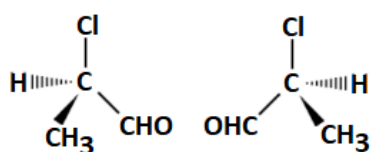
- For an alkene-based compound to exhibit stereoisomerism, there must be a minimum of 2 different functional groups at either end of the carbon=carbon double bond
 - Compound **1** has:
 - A hydrogen atom and a methyl group at one end of the C=C bond
 - A chlorine atom and a CH₂OH group at the other end of the C=C bond
 - Compound **4** has:
 - Two methyl groups at one end of the C=C bond
 - A hydrogen atom and a carboxylic acid group at the other end of the C=C bond
- The isomerism arises due to the π bond system above and below the σ bond formed between the carbon atoms
 - The π bond system means that there is no free rotation about that bond

1d

d)

i) The type of isomerism shown by compounds **2** and **3** is:

- Optical (isomerism); **[1 mark]**

ii) The **two** isomers of compound **2** are:

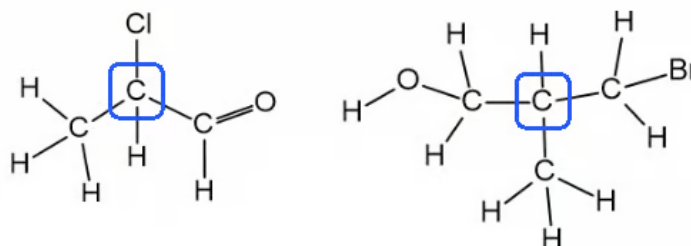
- **Two** correct isomers drawn
AND
Mirror image of each other; **[1 mark]**

This type of isomerism occurs because:

- There is a carbon atom that has four different atoms or group of atoms attached; **[1 mark]**
- So, there are two molecules that are non-superimposable mirror images of each other; **[1 mark]**

[Total: 4 marks]

- Chiral centres are carbon atoms with four different atoms or groups of atoms attached
- This causes optical isomerism as the four different atoms or groups of atoms can be drawn in two different orientations that are non-superimposable mirror images of one another
- Compounds **2** and **3** both have chiral centres in their structures



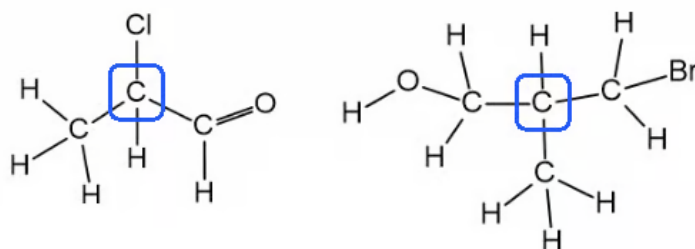
2a

a) The concept of hybridisation can be used to explain the bonding in the triple bond present in propyne as follows:

- The two carbon atoms are sp hybridised; [1 mark]
- Sigma bond formed from overlap of sp orbitals (of carbon atoms); [1 mark]
- Two pi bonds formed from overlap of unhybridised p orbitals (on each carbon atom); [1 mark]

[Total: 3 marks]

- A triple bond consists of one σ bond and two π bonds.
- To form a triple bond, the two carbon atoms must be sp hybridized
 - The σ bond forms from the overlap of an sp hybrid on each carbon atom
 - The two pi bonds are formed from the overlap of the two unhybridised p



2a

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- The two carbon atoms are sp hybridised; [1 mark]
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[Total: 3 marks]

- A triple bond consists of one σ bond and two π bonds.
- To form a triple bond, the two carbon atoms must be sp hybridized
 - The σ bond forms from the overlap of an sp hybrid on each carbon atom
 - The two pi bonds are formed from the overlap of the two unhybridised p orbitals on each carbon

2b

b) The number of carbon atoms in the molecule that exhibit sp^2 hybridisation is:

- 6; [1 mark]

[Total: 1 mark]

- Only the carbon atoms in double bonds will exhibit sp^2 hybridisation
- **Careful:** When you are rushing, it can be very easy to only count one of the carbons

in the double bond

- This happens because you are actually counting the double bonds and not the carbons

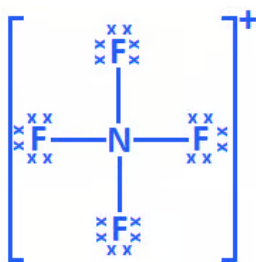
2c

c) The hybridisation shown by the nitrogen atoms in NF_4^+ , N_2H_2 and N_2H_4 is as follows:

	NF_4^+	N_2H_2	N_2H_4
Hybridisation	sp^3 ; [1 mark]	sp^2 ; [1 mark]	sp^3 ; [1 mark]

[Total: 3 marks]

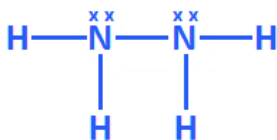
- We can use the number of electron domains around each atom to deduce the hybridisation
 - 4 electron domains = sp^3
 - 3 electron domains = sp^2
 - 2 electron domains sp
- The nitrogen in NF_4^+ has 4 bond pairs and no lone pairs so it has 4 electron domains



- Each nitrogen in N_2H_2 has 2 bond pairs and one lone pair so it has 3 electron domains



- Each nitrogen in N_2H_4 has 3 bond pairs and 1 lone pair so it has 4 electron domains



3a

a)

i) The IUPAC name of gasoline is:

- 2,2-dimethylpentane; [1 mark]

ii) The general formula for the homologous series to which gasoline belongs is:

- C_nH_{2n+2}; [1 mark]

[Total: 2 marks]

- The longest carbon chain is 5 carbons long
 - The IUPAC name will be pent- based
- There are only carbon-carbon single bonds
 - The IUPAC name will be pentane based
- There are 2 methyl groups on carbon-2
 - The IUPAC name will include 2,2-dimethyl
- The overall name is therefore 2,2-dimethylpentane
- The compound contains only carbon-carbon single bonds which means that it belong to the homologous series of alkanes
 - The general formula for the alkanes is C_nH_{2n+2}

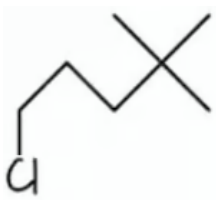
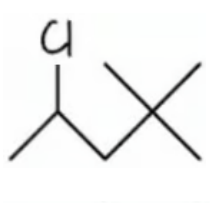
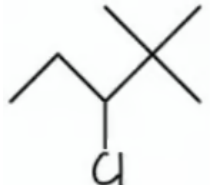
3b

b)

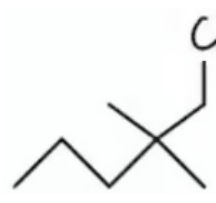
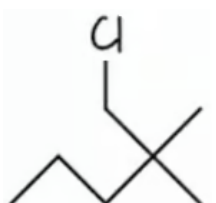
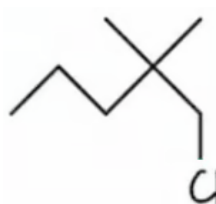
i) The type of organic mechanism used in this reaction is

- Free radical substitution; [1 mark]

ii) The skeletal formula of the **four** monochlorinated isomers formed are:

-  ; [1 mark]
-  ; [1 mark]
-  ; [1 mark]

Plus, any **one** of the following:

- 
OR

OR
 ; [1 mark]

iii) The type of isomerism shown by the isomers in (ii) is:

- Position / positional; [1 mark]

[Total: 6 marks]

- For part (i)
 - Alkanes are relatively unreactive

- The reagent and conditions mentioned in the question are typical of free radical substitution
- For part (ii)
 - You are told that the gasoline in Fig. 2.1 is being monochlorinated, which means that one hydrogen atom will be replaced with one chlorine atom
 - The question states to draw the skeletal formulae so make sure you don't draw any other representations
 - **Careful:** On the mark scheme, the first three isomers are all different but the last three are actually all the same
- For part (iii)
 - The possible isomers are 1-chloro-2,2-dimethylpentane, 3-chloro-2,2-dimethylpentane, 4-chloro-2,2-dimethylpentane and 5-chloro-2,2-dimethylpentane / 1-chloro-4,4-dimethylpentane
 - Since the chlorine atoms have moved / changed position, they are position / positional isomers

3c

c)

i) The role of the aqueous hydroxide ion in this reaction is:

- Nucleophile; [1 mark]

ii) The name of compound X is:

- Propan-1-ol; [1 mark]

iii) The reaction conditions and type of mechanism used in this alternative reaction are:

- Conditions: Heat / warm
AND

last three are actually all the same

- For part (iii)
 - The possible isomers are 1-chloro-2,2-dimethylpentane, 3-chloro-2,2-dimethylpentane, 4-chloro-2,2-dimethylpentane and 5-chloro-2,2-dimethylpentane / 1-chloro-4,4-dimethylpentane
 - Since the chlorine atoms have moved / changed position, they are position / positional isomers

3c

c)

i) The role of the aqueous hydroxide ion in this reaction is:

- Nucleophile; [1 mark]

ii) The name of compound X is:

- Propan-1-ol; [1 mark]

iii) The reaction conditions and type of mechanism used in this alternative reaction are:

- Conditions: Heat / warm

AND

Alcoholic / ethanolic (sodium) hydroxide; [1 mark]

- Mechanism: Elimination; [1 mark]

[Total: 4 marks]

- The hydroxide / OH^- ion is attacking a δ^+ carbon, which makes it a nucleophile
 - This is due to the difference in electronegativity of the carbon and bromine atoms in the C-Br bond
- The formation of a bromide ion / Br^- suggests that the Br and OH^- have switched places
 - This is a nucleophilic substitution reaction which will produce propan-1-ol
- You should be aware of the different reactions and mechanisms with a hydroxide ion

- Nucleophilic substitution to form an alcohol with aqueous sodium hydroxide
- Elimination to form an alkene and water when heated with alcoholic / ethanolic sodium hydroxide

3d

d)

i) In terms of orbitals and bonding, pentanal can be described as trigonal planar about the carbonyl carbon because:

- The carbonyl bond contains one sigma / σ bond; [1 mark]
- The p orbitals overlap to form a pi / π bond; [1 mark]
- The bonding pairs of electrons are in the same plane

AND

This causes repulsion; [1 mark]

ii) The shape and bond angle adopted by the carbonyl carbon after the reaction with hydrogen cyanide are:

- Tetrahedral

AND

109.5 ($^{\circ}$); [1 mark]

[Total: 4 marks]

- Saying that the carbonyl bond is sp^2 hybridised can achieve one of the first two marking points
- Any time that there is planarity within a molecule, there is some form of hybridisation in place
 - You should state about the sigma / σ bond as this, essentially forms the core of any covalent bond
 - You can then talk about the overlap of p orbitals to form a pi / π bond or system
 - This leads to some electrons being in the same plane causing repulsion
 - In the case of pentanal and other aldehydes, this means that the shape around the carbonyl carbon is trigonal planar with an angle of 120°
- When pentanal reacts with hydrogen cyanide, the product is 2-hydroxyhexanenitrile
 - **Careful:** The shape and angle that you are being asked for are for the 5th

carbon along in the chain (as you look at the product)

- The previously carbonyl carbon / fifth carbon now has four different atoms or groups of atoms attached which will adopt a tetrahedral shape with a bond angle of 109.5°
- A common incorrect answer is linear and 180° because students look at the end / nitrile carbon of the chain but this carbon came from the hydrogen cyanide

4a

a)

i) The shortest straight-chain alkane to have structural isomers and the isomer are:

- Shortest straight-chain alkane = butane; [1 mark]
- Isomer = 2-methylpropane; [1 mark]

ii) The type of isomerism shown by the isomers in part (i) is:

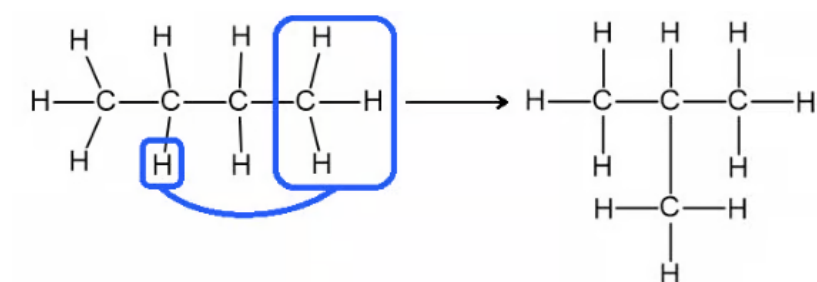
- Chain (isomerism)

OR

Branched chain / branch (isomerism); [1 mark]

[Total: 3 marks]

- The three different types of structural isomerism are:
 - Chain isomerism
 - Positional isomerism
 - Functional group isomerism
- It is not possible for a straight-chain alkane to exhibit any functional group isomerism because the molecule is saturated
 - This means that you are looking for chain or positional isomerism
- Methane, ethane and propane cannot have any chain or positional isomers because they are not long enough to snap a section off the main chain and place it elsewhere
- Butane is the first straight-chain alkane that is long enough to snap a section off
 - The section will be a methyl group
 - This leaves a chain of three carbons
 - The methyl group can replace a hydrogen on carbon-2 giving rise to 2-methylpropane



4b

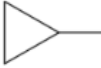
b)

i) The name of the homologous series that C_4H_8 belongs to is the:

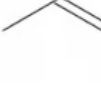
- Alkenes; [1 mark]

ii) The skeletal formulae of **two** cyclic C_4H_8 isomers and **two** C_4H_8 isomers that show cis / trans isomerism are:

Cyclic

-  ; [1 mark]
-  ; [1 mark]

Cis / trans isomerism

-  ; [1 mark]
-  ; [1 mark]

iii) Including explanations why they do not show cis / trans isomerism, the **two** unsaturated C_4H_8 isomers that do **not** show cis / trans isomerism are:

- But-1-ene

AND

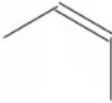
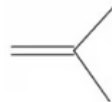
(Because,) one end of the carbon-carbon double bond / $C=C$ has two hydrogen atoms; [1 mark]

- Methylpropene

AND

(Because,) one end of the carbon-carbon double bond / $C=C$ has two methyl groups; [1 mark]

[Total: 7 marks]

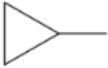
- The general formula for the alkenes homologous series is C_nH_{2n} , which matches with the C_4H_8 formula given in the question
- There are only two possible cyclic isomers with the formula C_4H_8
 - A cyclic ring containing four carbons, i.e. cyclobutane
 - A cyclic ring containing three carbons, i.e. cyclopropane, but then replacing one of the hydrogens with a methyl group gives you the second cyclic isomer with the formula C_4H_8
- There are four possible unsaturated isomers with the formula C_4H_8
 - Some of these will show cis / trans isomerism, some will not
- **Remember:** Cis / trans isomerism occurs when there are functional groups on the same or opposite sides of the double bond as a result of the lack of free rotation about the carbon-carbon double bond / $C=C$
 - **Tip:** It can help to draw out all of the possibilities and then you can decide which isomers do or don't show cis / trans isomerism
 -  - does not show cis / trans isomerism as the left-hand end of the $C=C$ bond has two hydrogen atoms
 -  - does show cis / trans isomerism
 -  - does show cis / trans isomerism
 -  - does not show cis / trans isomerism as the left-hand end of the

i) The name of the homologous series that C_4H_8 belongs to is the:

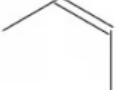
- Alkenes; [1 mark]

ii) The skeletal formulae of **two** cyclic C_4H_8 isomers and **two** C_4H_8 isomers that show cis / trans isomerism are:

Cyclic

-  ; [1 mark]
-  ; [1 mark]

Cis / trans isomerism

-  ; [1 mark]
-  ; [1 mark]

iii) Including explanations why they do not show cis / trans isomerism, the **two** unsaturated C_4H_8 isomers that do **not** show cis / trans isomerism are:

- But-1-ene

AND

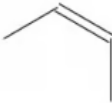
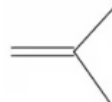
(Because,) one end of the carbon-carbon double bond / $C=C$ has two hydrogen atoms; [1 mark]

- Methylpropene

AND

(Because,) one end of the carbon-carbon double bond / $C=C$ has two methyl groups; [1 mark]

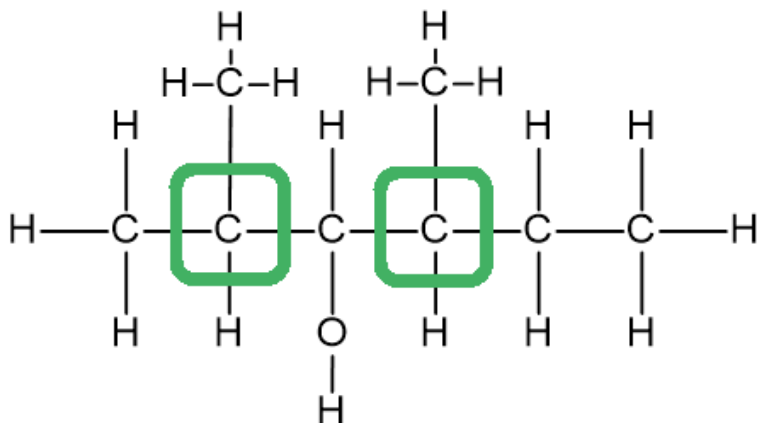
[Total: 7 marks]

- The general formula for the alkenes homologous series is C_nH_{2n} , which matches with the C_4H_8 formula given in the question
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 - **Tip:** It can help to draw out all of the possibilities and then you can decide which isomers do or don't show cis / trans isomerism
 -  - does not show cis / trans isomerism as the left-hand end of the $C=C$ bond has two hydrogen atoms
 -  - does show cis / trans isomerism
 -  - does show cis / trans isomerism
 -  - does not show cis / trans isomerism as the left-hand end of the $C=C$ has two hydrogen atoms and the right-hand end of the $C=C$ bond has two methyl groups

4c

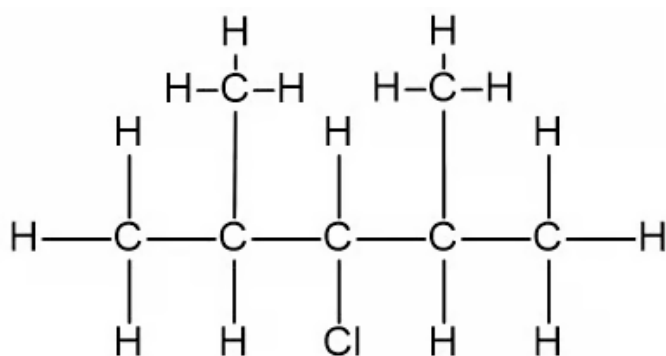
c) The displayed formulae and, where appropriate, any chiral centres are:

2,4-dimethylhexan-3-ol



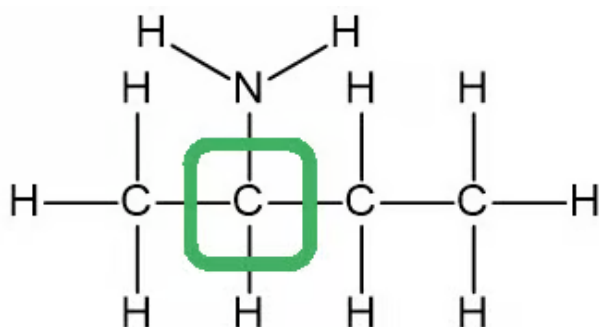
- Correct displayed formula including the O-H bond; [1 mark]
- Chiral centre on carbon-2 identified; [1 mark]
- Chiral centre on carbon-4 identified; [1 mark]

2,4-dimethyl-3-chloropentane



- Correct displayed formula; [1 mark]

2-aminobutane



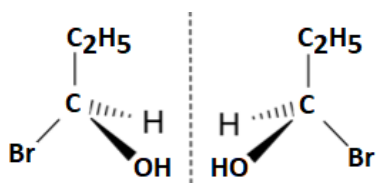
- Correct displayed formula including the N-H bonds; [1 mark]
- Chiral centre identified; [1 mark]

[Total: 6 marks]

- **Remember:** Chiral centres are carbon atoms with four different atoms or groups of atoms attached
- **Careful:** When you are asked for displayed formulae, you should show **all** of the bonds
 - You will sometimes have mark schemes that allow things like OH and NH₂ rather than having the bonds displayed

4d

d) 3D representations of the **two** enantiomers of 1-bromopropan-1-ol are:



- Correct structure of one enantiomer; [1 mark]
- Second enantiomer shown as a mirror image; [1 mark]

[Total: 2 marks]

- Other 3D representations are possible / acceptable as it will depend on where you place each of the four different atoms / groups of atoms in the first enantiomer
 - For 1-bromopropan-1-ol, the chiral carbon should have a hydrogen atom, bromine atom, hydroxyl group and ethyl group attached
- The second enantiomer should always be a mirror image of the first one

5a

a) Cyclohexane is non-planar because:

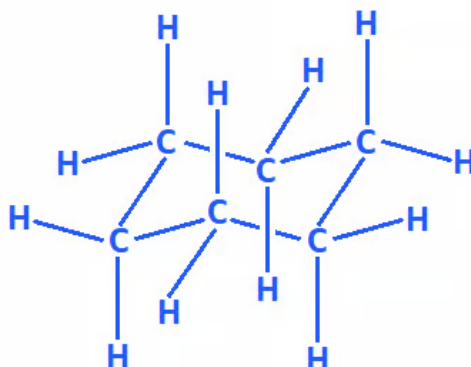
- The carbon atoms are sp³ hybridised; [1 mark]
- (Each forming) a tetrahedral arrangement around each carbon
AND
(The) bond angles of 109.5° (give the puckered shape); [1 mark]

[Total: 2 marks]

- Hybridisation can be used to explain the shape of any molecule
 - Conversely, the shape of a molecule can be used to determine the type of

hybridisation that has occurred.

- Tetrahedral arrangement $\leftrightarrow$ sp^3 hybridised
- Triangular planar arrangement $\leftrightarrow$ sp^2 hybridised
- Linear arrangement $\leftrightarrow$ sp hybridised
- The Lewis structure for cyclohexane can help us to see how the tetrahedral arrangement around each carbon creates the puckered shape



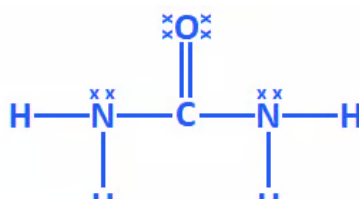
5b

b) The hybridisation of C and N in urea and approximate bond angles are:

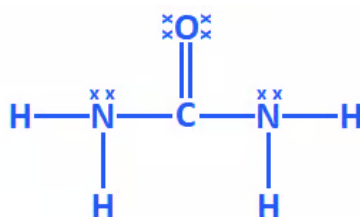
- Carbon is sp^2 hybridised; [1 mark]
- With a bond angle of 120° ; [1 mark]
- Nitrogen is sp^3 hybridised; [1 mark]
- With a bond angle of 109.5° ; [1 mark]

[Total: 4 marks]

- It can be useful to start with the Lewis (electron dot) structure for the molecule



- It can be useful to start with the Lewis (electron dot) structure for the molecule



- Looking at carbon, we can see it has 3 electron domains
- Looking at nitrogen, we can see it has 4 electron domains
- VSEPR theory, applied to hybridised orbitals in the same way as to electron domains, explains the geometry of the hybrid atomic orbitals
 - It, therefore, relates the shape of the molecule to the hybridisation and to the number of electron domains
 - Number of electron domains $\leftrightarrow$ shape of molecule $\leftrightarrow$ hybridisation

5c

c) The hybridisation of the carbon atom in methane is

- sp^3 ; [1 mark]
- The isolated carbon atom electron configuration $1s^2 2s^2 2p^2$ / excited state excited electron configuration is $1s^2 2s^1 2p^3$

AND

$2s^1 2p^3$ electrons blend to form four identical hybrid orbitals; [1 mark]

- Hybrid orbitals are lower in total energy than atomic orbitals; [1 mark]
 - Repulsion of (identical hybrid) orbitals creates a tetrahedral shape; [1 mark]
- (A suitably annotated diagram is also acceptable for the marks)

[Total: 4 marks]

- Carbon has the following electron configuration
 - $1s^2 2s^2 2p^2$
- One of the electrons in the 2s orbital of the ground-state configuration of carbon is promoted to the vacant 2p orbital
- The four atomic orbitals 2s, $2p_x$, $2p_y$, and $2p_z$ combine to form a set of four new hybrid orbitals.
- The four new sp^3 hybrid orbitals are entirely equivalent
- Each one consists of 25% s character and 75% p character since they were formed from one s orbital and three p orbitals.

- The shape of each sp^3 hybrid orbital will have 75% of the characteristics of a p orbital (dumbbell-shaped) combined with 25% of the characteristics of the spherical s orbital
- Each carbon sp^3 orbital with a hydrogen $1s$ atomic orbital
- Hydrogen has a $1s^1$ electron configuration and the s orbital is spherical
- Any molecule with a tetrahedral electron domain geometry on its interior central atom (based on four electron domains) would be predicted to have sp^3 hybridisation

5d

d) The type of hybridisation exhibited by the carbon atom marked is:

- sp^3 ; [1 mark]

[Total: 1 mark]

- Each carbon atom has made four single bonds
- Therefore, all those bonds are sigma bonds and no lone pairs exist on both carbon atoms
- In fact, both carbon atoms and the oxygen atom exhibit sp^3 hybridisation in this molecule

